Lecture Notes: Integration of Quantum Computing, AI, and HPC for Accelerating Materials Discovery

Speaker: Dr. Seongmin Kim (Oak Ridge National Laboratory)
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1. Introduction and Research Journey

Dr. Kim's research focuses on an integrated framework leveraging **Quantum Computing (QC), Artificial Intelligence (AI/ML), and High-Performance Computing (HPC)** to solve difficult materials science problems.

- **The Limitations of Experimental Science:** Dr. Kim began his career focusing on experimental materials science (e.g., hydrophobic materials, anti-wetting bio-tubes). However, he found that relying on physical trial-and-error experimentation was incredibly slow and resource-intensive. There are simply too many combinations of materials, geometries, and fabrication conditions to explore manually.
- **The AlphaGo Inspiration:** Inspired by AlphaGo's ability to navigate massive design spaces smartly, Dr. Kim shifted toward computational materials science to make the discovery process more efficient.
- **The Exponential Problem:** While machine learning is excellent for predicting material performance, optimization becomes impossible for classical computers as variables increase. For example, a 50-pixel metamaterial yields 2^{50} possible configurations, and adding just one pixel doubles the design space to 2^{51}. This exponential growth led to the adoption of quantum computing.

2. Quantum Computing in Materials Science

To bridge materials optimization and quantum computing, materials must be translated into a format the quantum computer understands.

- **Binary Representation:** Material candidate structures are represented as binary bit strings (e.g., 0s and 1s indicating the absence or presence of a material feature).
- **The Hamiltonian Energy Landscape:** In physics, a Hamiltonian describes the energy of a system. In this framework, the Hamiltonian acts as an energy landscape covering all possible binary designs, where the **lowest energy state (the ground state) represents the optimal material design**.

3. Bridging ML and QC: The Surrogate Model

Because testing every possible material to find its exact energy is prohibitively expensive, researchers only have sparse datasets.

- **Factorization Machines:** To fill the gaps, Dr. Kim uses a supervised machine learning model called a Factorization Machine (FM). FMs efficiently capture linear terms and pairwise interactions between variables.
- **Mapping to QUBO:** The linear and quadratic coefficients produced by the FM are mapped directly to a **Quadratic Unconstrained Binary Optimization (QUBO)** format. This QUBO acts as a surrogate Hamiltonian that can be directly optimized by a quantum computer.
- **Inverse Design:** Instead of predicting the performance of a known structure, this framework performs "inverse design," starting from the target objective and using the quantum optimizer to find the binary vector (material structure) that achieves it.

4. The Active Learning Loop

Because the initial dataset used to train the QUBO might only contain 25 to 50 data points, the surrogate model's first predictions will be imperfect.

- To refine it, the workflow uses an **active learning feedback loop**.
- The QC predicts an optimal structure based on the current QUBO, which is then validated using accurate physics-based simulations (like finite element methods).
- This validated result is added back into the dataset, the FM is retrained, and the QUBO becomes progressively more accurate over multiple iterations.

5. Quantum Hardware: Annealing vs. Gate-Based

Dr. Kim outlines two ways to solve QUBOs:

- **Quantum Annealing:** Highly specialized for finding low-energy states of objective functions like QUBOs. It maps easily to ML surrogate models and can be used in hybrid setups for large problems. However, it is not a universal computer and suffers from restricted connectivity, embedding overhead, and hardware noise.
- **Gate-Based Quantum Computing (QAOA):** A universal, flexible approach using parameterized quantum circuits, solved via the **Quantum Approximate Optimization Algorithm (QAOA)**.
- **The NISQ Challenge:** We are currently in the Noisy Intermediate-Scale Quantum (NISQ) era. QAOA struggles because as problems grow, circuit depth increases, causing gate errors to accumulate rapidly until the result is unreliable. Furthermore, simulating large quantum circuits on classical computers is impossible due to exponential memory requirements.

6. Distributed QAOA (DQAOA) & Quantum-Centric Supercomputing

Because standard QAOA struggles with problems larger than 30 variables, Dr. Kim introduced **DQAOA**.

- **Decomposition:** DQAOA breaks a massive QUBO down into smaller, manageable sub-problems.
- **HPC Integration:** These sub-problems are solved in parallel on a hybrid HPC-QC system and iteratively aggregated into a global solution.
- **Scale:** DQAOA dramatically scales the capabilities of quantum optimization, successfully handling problems with up to 10,000 variables. Furthermore, integrating GPUs into the classical portion of the workflow speeds up the process by more than 10 times.

7. Real-World Applications

The framework has been proven on several practical applications:

- **Transparent Radiative Coolers:** Designed layered windows that passively control light and heat (high visible transmission, low UV/NIR, strong IR emission) to reduce cooling energy in buildings.
- **High-Entropy Alloys:** Applied the DQAOA framework to compositionally complex metals, using Machine Learning Potentials (neural networks) as high-fidelity surrogates to avoid slow DFT/MD calculations.
- **Radio-Frequency Integrated Circuits (RFICs):** Proved the materials framework is highly flexible and can be repurposed to optimize electronics and communications hardware.

8. Advanced Formulations and Formats

To push toward highly realistic, messy scientific problems, the framework uses advanced adaptations:

- **DVQOA (Distributed Variational Quantum Optimization Algorithm):** An unsupervised alternative to QAOA that acts similarly to a neural network. It interacts directly with the objective function, bypassing the mandatory QUBO/Hamiltonian conversion step.
- **Continuous Variables:** Real materials use continuous variables (like layer thickness) alongside discrete binary choices. The framework discretizes these continuous variables into binary bit strings, solves them combined in the quantum hardware, and decodes them back into continuous metrics.

- **HUBOs (Higher-Order Unconstrained Binary Optimization):** While QUBOs only model 2-variable pairwise interactions, realistic physics requires modeling 3+ interacting variables simultaneously. HUBOs achieve this accurately but make the energy landscape extremely rugged and hard to solve. DQAOA and HPC are mandatory to navigate this rugged HUBO landscape efficiently.

9. QAOA-GPT: Generative AI for Circuit Design

To skip the slow, iterative tuning loop of variational algorithms like QAOA, Dr. Kim introduced **QAOA-GPT**.

- A generative AI model learns what optimal quantum circuits look like and instantly generates them for new problems (inference takes less than 0.1 seconds).
- While training the GPT requires generating expensive, high-quality data, **DQAOA perfectly mitigates this weakness**. Because DQAOA relies on solving small, fixed-size sub-problems, the GPT only needs to be trained on small datasets, yielding up to a 27x speedup over standard DQAOA.

10. Future Work and Hardware Utilization

Looking forward, Dr. Kim highlighted a major priority for efficiently using quantum chips.

- **Simultaneous Execution:** Rather than running one DQAOA sub-problem at a time, researchers can combine independent sub-problems into a single Hamiltonian.
- This allows a single quantum device to process multiple sub-problems simultaneously. Critically, this increases circuit width (using more qubits) while **keeping the circuit depth constant**, which is essential for mitigating errors on current NISQ hardware. This also prepares the framework for future Fault-Tolerant Quantum Computing.

11. Conclusion and Q&A Highlights

Dr. Kim concluded that quantum computing is no longer just theoretical—it is an active tool for doing real science when deeply integrated with HPC and AI.

- **Dominant Error Source:** During the Q&A, Dr. Kim clarified that accumulated noise from deep quantum circuits on physical hardware is currently the largest source of error, alongside poorly constructed classical datasets.
- **Classical/Quantum Boundary:** The quantum computer does not process classical data directly; it only evaluates the QUBO Hamiltonian generated by the classical AI.

- **Learning Roadmap:** For beginners, Dr. Kim recommends starting with Quantum Annealing due to its approachability, followed by learning gate-based QAOA.